RIEMANN-ROCH THEOREM

In this lecture, we continue our discussion of divisors on smooth projective curves which culminates in the Riemann-Roch Theorem. As before, we will not prove all the things covered here, since these lectures are to be taken partly as a glossary of terms and results that we will use. The students are expected to work out omitted details on their own, as this will give them much needed experience with these objects.

1. More properties of divisors

As before, we will assume C to be a smooth curve.

Definition 1.1. Let $D = \sum_{P \in C} n_P P$ be a divisor on C. Its *support* is

$$\mathrm{Supp}(D) = \{P \in C : n_P \neq 0\}.$$

We say that D is *effective*, and write $D \geq 0$, if $n_P \geq 0$ for every P. More generally, for two divisors D and E, we write $D \geq E$ if $D - E$ is effective. Every divisor has a unique decomposition $D = D^+ - D^-$, where

$$D^+ = \sum_{n_P > 0} n_P P, \qquad D^- = \sum_{n_P < 0} (-n_P) P$$

are effective divisors with disjoint support.

Proposition 1.2. *Let D be a divisor on C.*

(1) If $D \geq 0$ and $\deg D = 0$, then $D = 0$.
(2) If D is principal and effective, then $D = 0$.

Proof. The first assertion is immediate from

$$\deg D = \sum_{P \in C} n_P$$

and the fact that all n_P are nonnegative. The second follows from the first since a principal divisor has degree zero. (Equivalently, a rational function with no poles on a projective curve is constant.) □

Divisors allow us to prescribe where a rational function is allowed to have poles and where it is required to have zeroes.

Definition 1.3. For a divisor D on C, define

$$L(D) := \{f \in \bar{K}(C)^\times : \mathrm{div}(f) + D \geq 0\} \cup \{0\}.$$

This is a vector space over $\bar{K}$. We write

$$\ell(D) = \dim_{\bar{K}} L(D).$$

Note that *a priori* $\ell(D)$ is not known to be finite, but the next proposition will show that it is in fact a finite number.

Thus, if $D = \sum_P n_P P$, then a nonzero function f belongs to $L(D)$ precisely when

$$\mathrm{ord}_P(f) \geq -n_P$$

for every P. In particular, a positive coefficient n_P allows a pole of order at most n_P, while a negative coefficient forces a zero of order at least $-n_P$.

Proposition 1.4 (First properties of $L(D)$)**.** *Let D and E be divisors on C.*

(1) If $D \leq E$, then $L(D) \subseteq L(E)$.
(2) $L(0) = \bar{K}$.
(3) If $\deg D < 0$, then $L(D) = 0$.

(4) If $D' = D + \operatorname{div}(h)$ for some $h \in \bar{K}(C)^\times$, then

$$L(D) \longrightarrow L(D'), \qquad f \longmapsto \frac{f}{h},$$

is an isomorphism. In particular, $\ell(D)$ depends only on the linear equivalence class of D.

(5) The space $L(D)$ is finite dimensional. More precisely,

$$\ell(D) \leq \max\{0, \deg D + 1\}.$$

Proof. The first assertion follows directly from the definition. The second follows from the fact that a rational function on a projective curve with no poles is constant.

If $0 \neq f \in L(D)$, then

$$\operatorname{div}(f) + D \geq 0.$$

Taking degrees gives

$$\deg D = \deg\big(\operatorname{div}(f) + D\big) \geq 0,$$

which proves the third assertion.

For the fourth assertion, if $f \in L(D)$, then

$$\operatorname{div}(f/h) + D' = \operatorname{div}(f) - \operatorname{div}(h) + D + \operatorname{div}(h) = \operatorname{div}(f) + D \geq 0.$$

The inverse map is multiplication by h.

It remains to prove finite dimensionality and the stated bound. We may suppose $L(D) \neq 0$, and choose $0 \neq f_0 \in L(D)$. Then

$$E = D + \operatorname{div}(f_0)$$

is effective and linearly equivalent to D, so $\ell(D) = \ell(E)$ and $\deg E = \deg D$.

Write $d = \deg E$, and choose a chain of effective divisors $0 = E_0 \leq E_1 \leq \cdots \leq E_d = E$ such that $E_i = E_{i-1} + P_i$ for some point P_i, where points may be repeated. Suppose that the coefficient of P_i in E_i is m. If t is a uniformizer at P_i, then the map

$$L(E_i) \longrightarrow \bar{K}, \qquad f \longmapsto (t^m f)(P_i),$$

has kernel $L(E_{i-1})$. Hence $\dim_{\bar{K}} L(E_i) - \dim_{\bar{K}} L(E_{i-1}) \leq 1$. Since $L(0) = \bar{K}$, induction gives

$$\ell(E) \leq d + 1.$$

This proves the result. □

Corollary 1.5. *Let D be a divisor of degree zero. Then $\ell(D) > 0$ if and only if D is principal. In this case $\ell(D) = 1$.*

Proof. Suppose $0 \neq f \in L(D)$. Then $\operatorname{div}(f) + D$ is an effective divisor of degree zero, and is therefore zero. Hence $D = -\operatorname{div}(f)$ is principal. Conversely, if $D = -\operatorname{div}(f)$, then $f \in L(D)$. Moreover, if $g \in L(D)$, then

$$\operatorname{div}(g/f) \geq 0,$$

so g/f is constant. Thus $L(D) = \bar{K} f$. □

We can apply similar ideas to divisors of differentials as well.

Definition 1.6. For a divisor D on C, define:

$$\Omega(D) = \{\omega \in \Omega(C) : \operatorname{div}(\omega) \geq D\} \cup \{0\}.$$

We write $i(D) = \dim_{\bar{K}} \Omega(D)$. In particular, $\Omega(0) = \{\text{regular differentials on } C\} \cup \{0\}$. The *genus* of C is $g(C) = \ i(0) = dim_{\bar{K}}\Omega(0)$.

Now we begin to explore the connection between functions and differentials on curves at a deeper level.

Proposition 1.7. *Fix a nonzero differential ω_0 on C and let $K_C = \operatorname{div}(\omega_0)$. Then multiplication by ω_0 gives an isomorphism $L(K_C - D) \cong \Omega(D)$. Consequently, $i(D) = \ell(K_C - D)$, and $\Omega(D)$ is finite dimensional.*

Proof. We have $f\omega_0 \in \Omega(D)$ if and only if $\operatorname{div}(f) + K_C \geq D$, which is equivalent to

$$\operatorname{div}(f) + (K_C - D) \geq 0.$$

This is precisely the condition that $f \in L(K_C - D)$. □

2. Riemann-Roch and Riemann-Hurwitz theorem

Next theorem is a fundamental result in the algebraic geometry of curves. It tells us 'how many functions are there on C having prescribed zeroes and poles'.

Theorem 2.1 (Riemann-Roch)**.** *Let C be a smooth curve and let K_C be a canonical divisor on C. Then for every divisor $D \in \mathrm{Div}(C)$,*

$$\ell(D) - \ell(K_C - D) = \deg D - g + 1.$$

Remark 2.2. We will not prove this theorem, and the classical proof is not very interesting. However, note the following way of writing this formula:

$$\ell(D) - \ell(K_C - D) + \ell(K_C) = \deg D + 1.$$

Note the symmetry when it is written this way. The alternating sum on LHS is a an alternating sum of dimensions of vector spaces associated with D and C while the RHS is a geometric quantity associated with D. This *should* feel like some magic. Indeed, such an arrangement is indicative of a 'trace formula' (in this case, the more general Hirzebruch-Riemann-Roch theorem), or an 'index theorem' working in the background; which are some of the most powerful tools you will see in algebraic geometry and arithmetic geometry. You should always look out for them in order to obtain and enhance understanding.

Corollary 2.3. *(i)* $\deg K_C = 2g - 2$.
(ii) If $\deg D > 2g - 2$, *then* $\ell(D) = \deg D - g + 1$.

The next theorem is a statement about tame ramification and wild ramification in disguise. But we don't need to worry about those terms at least as of now. We will have occasion to use the formula on times. But you should think about ramification and its properties while trying to prove the theorem.

Theorem 2.4 (Riemann-Hurwitz)**.** *Let $\phi : C_1 \to C_2$ be a nonconstant separable morphism of smooth curves of genera g_1 and g_2 respectively. Then:*

$$2g_1 - 2 \geq (\deg \phi)(2g_2 - 2) + \sum_{P \in C_1} (e_\phi - 1).$$

Moreover, equality holds if and only if: either $\mathrm{char}K = 0$, *or* $\mathrm{char}K = p > 0$ *and p does not divide $e_\phi(P)$ for all $P \in C_1$.*

3. Galois descent

In this section, we see that if a curve and divisor are defined over a non-algebraically closed field K, then $L(D)$ has a basis consisting of functions in $K(C)$. This is a 'Galois descent' statement, which we state as the following lemma.

Lemma 3.1. *Let V be a $\bar{K}$-vector space. (Not necessarily finite dimensional!) Assume G_K acts on V continuously and compatibly with its action on $\bar{K}$. Then*

$$V \cong \bar{K} \otimes_K (V^{G_K})$$

where V^{G_K} denotes the K-vector space of elements in V fixed by every $\sigma \in G_K$.

This is the same as saying that a basis for V can be chosen consisting of G_K-invariant vectors.

The proof of this lemma is a tricky exercise in field theory, but quite fundamental. We leave this as an exercise. From the lemma, the following proposition follows in a straightforward fashion.

Proposition 3.2. *Let C/K be a smooth curve and let $D \in \mathrm{Div}(C)$. Then $L(D)$ has a basis consisting of functions in $K(C)$.*